

**2024 WASSCE PC ECONOMICS 2 SOLUTION
PAPER 2**

SECTION A

Question 1

(a) Rate of growth of real GNP in both countries.

The formula for the rate of growth is: $((\text{New Value} - \text{Old Value}) / \text{Old Value}) \times 100\%$

- **Country X:**

- New GNP (2015) = \$75,000,000
- Old GNP (2010) = \$48,000,000
- Growth Rate = $\frac{75,000,000 - 48,000,000}{48,000,000} \times 100\%$
 $= \frac{27,000,000}{48,000,000} \times 100\%$
 $= \mathbf{56.25\%}$

- **Country Y:**

- New GNP (2015) = \$90,000,000
- Old GNP (2010) = \$54,000,000
- Growth Rate = $\frac{90,000,000 - 54,000,000}{54,000,000} \times 100\%$
 $= \frac{36,000,000}{54,000,000} \times 100\%$
 $= \mathbf{66.67\%}$

(b) Per capita income in both countries in 2010.

The formula for per capita income is: $\text{Real GNP} / \text{Population}$

- **Country X (2010):**

- Per Capita Income = $\frac{48,000,000}{18,400,000}$
 $= \mathbf{\$2.61}$

- **Country Y (2010):**

- Per Capita Income = $\frac{54,000,000}{9,450,000}$
 $= \mathbf{\$5.71}$

(c) Which country enjoyed a better standard of living in 2010?

Country Y enjoyed a better standard of living in 2010.

- **Explanation:** Per capita income is a primary indicator of the standard of living. It represents the average income per person in a country. In 2010, Country Y had a per capita income of \$5.71, which is more than double Country X's per capita income of \$2.61. This suggests that, on average, citizens of Country Y had more income available for goods and services, indicating a higher material standard of living.

(d) State of savings and investment in Country Y.

Based on the higher per capita income in Country Y (\$5.71):

- **(i) Level of savings:** The level of savings is likely to be **relatively high**. With a higher average income, individuals can cover their basic consumption needs and still have a surplus left over for savings.
- **(ii) Level of investment:** The level of investment is also likely to be **relatively high**. A higher level of domestic savings provides the capital base necessary for investment in new businesses, technology, and infrastructure, fostering economic growth.

(e) Net income from abroad and real GDP for Country Y in 2015.

- **Net Income from Abroad:**
 - Formula: Factor income from abroad - Factor income paid to foreigners
 - Net Income = \$14,800,000 - \$8,420,000 = **\$6,380,000**
- **Real Gross Domestic Product (GDP) in 2015:**
 - Formula: Real GNP - Net Income from Abroad
 - Real GDP = \$90,000,000 - \$6,380,000 = **\$83,620,000**

Question 2

(a) Income elasticity of demand for wheat.

The formula for income elasticity of demand (YED) is: (% Change in Quantity Demanded) / (% Change in Income)

- **(i) Engineer:**
 - Old Income (I_1) = \$3,000; New Income (I_2) = \$5,000
 - Old Quantity (Q_1) = 15 kg; New Quantity (Q_2) = 10 kg
 - $$YED = \frac{((10 - 15) / 15)}{((5000 - 3000) / 3000)}$$
$$= (-0.333) / (0.667)$$
$$= -0.5$$
- **(ii) Nurse:**
 - Old Income (I_1) = \$2,500; New Income (I_2) = \$4,000
 - Old Quantity (Q_1) = 8 kg; New Quantity (Q_2) = 12 kg
 - $$YED = \frac{((12 - 8) / 8)}{((4000 - 2500) / 2500)}$$
$$= (0.5) / (0.6)$$
$$= 0.83$$

(b) What type of good is wheat?

- **(i) Engineer:** For the Engineer, wheat is an **inferior good**.
- **(ii) Nurse:** For the Nurse, wheat is a **normal good** (specifically, a necessity).

(c) Reason for your answers.

- **(i) Engineer:** The income elasticity of demand is negative (-0.5). This means that as the Engineer's income increased, his demand for wheat decreased. Inferior goods are goods for which demand falls as consumer income rises.
- **(ii) Nurse:** The income elasticity of demand is positive but less than one (0.83). This means that as the Nurse's income increased, her demand for wheat also increased, but by a smaller proportion. This is characteristic of a normal good that is a necessity.

(d) System of taxation employed.

- **(i) Engineer:**
 - **System: Regressive taxation.**
 - **Reason:** The tax rate decreased as his income increased. The tax rate on his old (lower) income was 15%, while the tax rate on his new (higher) income was 10%. A regressive tax system is one where the tax rate decreases as the taxable amount increases.
- **(ii) Nurse:**
 - **System: Progressive taxation.**
 - **Reason:** The tax rate increased as her income increased. The tax rate on her old (lower) income was 8%, while the tax rate on her new (higher) income was 10%. A progressive tax system is one where the tax rate increases as the taxable amount increases.

SECTION B

Question 3

(a) Explanation of economic concepts:

- **(i) Production:** This is the process of creating or transforming inputs (like land, labor, capital, and entrepreneurship) into outputs, which are goods and services that have value and can satisfy human wants. It involves activities ranging from agriculture and manufacturing to providing services like healthcare and education.
- **(ii) Distribution:** This refers to the way in which the total output, income, or wealth of a nation is shared among its population. It addresses the question of "For whom are the goods and services produced?" and involves the allocation of income to the factors of production (wages for labor, rent for land, interest for capital, and profit for entrepreneurship).
- **(iii) Consumption:** This is the final use of goods and services by individuals, households, or institutions to satisfy their wants and needs. It is the ultimate goal of all economic activity. When a good or service is consumed, it is either used up or its value is diminished.

(b) Relationship between concepts:

- **(i) Production and Consumption:** These two concepts are fundamentally linked and interdependent. Production is undertaken to meet the demands of consumers; without consumption, there would be no reason to produce. Conversely, consumption is limited by what is produced; you cannot consume what does not exist. The level and pattern of consumption directly influence what and how much is produced.
- **(ii) Consumption and Distribution:** The distribution of income directly determines the level and pattern of consumption. Individuals or households with higher incomes have a greater ability to consume more goods and services than those with lower incomes. Therefore, how wealth and income are distributed in a society shapes its overall consumption patterns. An unequal distribution may lead to high consumption of luxury goods by a few, while a more equal distribution may lead to higher mass consumption of basic goods.

Question 4

(a) Definition of Tax:

A tax is a compulsory financial charge or levy imposed by a government on an individual, property, or economic activity (like a transaction) to fund government spending and various public expenditures.

(b) Features of a good tax system:

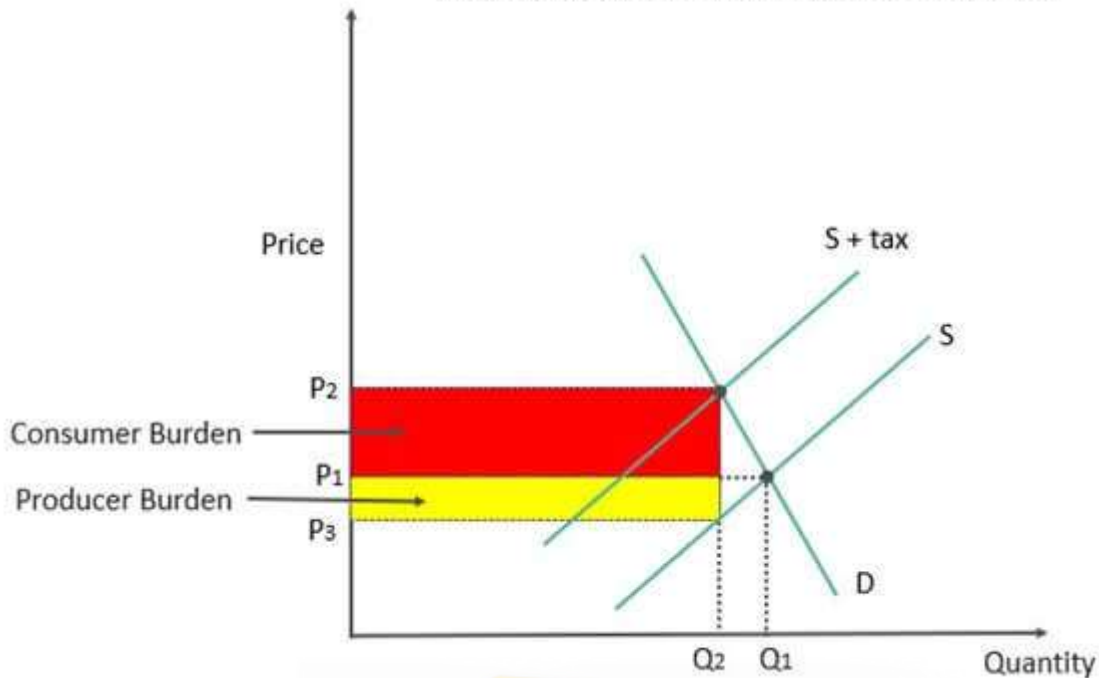
- **(i) Certainty:** The taxpayer should know with certainty the amount of tax to be paid, the time of payment, and the method of payment. This predictability is crucial for both individuals and businesses to plan their finances effectively and prevents arbitrary demands from tax officials.
- **(ii) Convenience:** The process of paying the tax should be convenient for the taxpayer. This means the timing and method of payment should be designed to cause the least amount of trouble. For example, income tax is often deducted at the source (Pay-As-You-Earn), which is convenient for the employee.

(c) Tax incidence with elastic demand:

If the demand for a good is elastic (elasticity > 1), it means consumers are very responsive to price changes. A small increase in price leads to a large decrease in quantity demanded.

When a tax is imposed on such a good, the **seller (producer) will bear a larger share of the tax burden**, while the **buyer (consumer) will bear a smaller share**.

Tax Incidence with Inelastic Demand



Question 5

(a) Definition of Value of Money:

The value of money refers to its purchasing power—the quantity of goods and services that a unit of money can buy. There is an inverse relationship between the value of money and the general price level. When prices are high (inflation), the value of money is low; when prices are low (deflation), the value of money is high.

(b) Distinction between commodity money and token money:

- **Commodity Money:** This is a type of money that has value outside of its use as money. Its value comes from the commodity of which it is made. Examples include gold, silver, salt, and cattle. The intrinsic value (the value of the material itself) is equal to its face value (its value as money).
- **Token Money:** This is a type of money whose face value is greater than its intrinsic value (the value of the material it is made from). Modern coins and paper currency are examples. The cost to produce a \$100 bill is far less than \$100. Its value is based on the trust and confidence people have in the issuing authority (the government or central bank).

(c) How the following can cause inflation:

Inflation is a sustained increase in the general price level of goods and services in an economy over a period of time.

- **(i) Velocity of circulation of money:** This is the rate at which money changes hands in an economy. If the velocity of circulation increases, it means people are spending money more quickly. Even if the supply of money and the volume of goods remain constant, this increased spending will drive up demand for goods and services faster than they can be supplied, leading to demand-pull inflation.
- **(ii) The supply of money:** If the central bank increases the supply of money (e.g., by printing more currency) at a rate faster than the growth in the output of goods and services, there will be more money chasing the same amount of goods. This excess money supply reduces the value of money and leads to a general rise in prices (demand-pull inflation).
- **(iii) Volume of goods and services:** If the volume (supply) of goods and services in an economy decreases while the money supply and velocity of circulation remain constant, there will be the same amount of money chasing fewer goods. This scarcity will drive up prices, leading to cost-push inflation. This can happen due to factors like natural disasters, wars, or supply chain disruptions.

Question 6

(a) Definition of a Commercial Bank:

A commercial bank is a type of financial institution that accepts deposits from the general public, creates credit, and provides various financial services such as making loans, processing payments, and offering savings and checking accounts to individuals and businesses. Their primary objective is to make a profit.

(b) Three functions of the Central Bank:

1. **Banker to the Government:** The Central Bank acts as the main financial agent for the government. It manages the government's main bank accounts, collects taxes and other receipts on its behalf, and makes payments for government expenditures. It also manages public debt and advises the government on monetary and financial matters.
2. **Issuer of Currency:** The Central Bank has the sole authority to issue a country's legal tender (notes and coins). This ensures uniformity in the currency and allows the bank to control the money supply.
3. **Banker to Commercial Banks and Lender of Last Resort:** The Central Bank holds cash reserves of commercial banks and provides clearinghouse facilities for the settlement of inter-bank claims. Crucially, it acts as the "lender of last resort" by providing loans to commercial banks that are facing liquidity problems and cannot get funds from elsewhere, thereby preventing banking crises.

(c) Instruments of monetary control:

These are tools used by the Central Bank to regulate the money supply, credit availability, and interest rates to achieve macroeconomic objectives like controlling inflation and promoting economic growth.

- **(i) Bank Rate (or Discount Rate):** This is the interest rate at which the Central Bank lends money to commercial banks. If the Central Bank wants to reduce the money supply (to fight inflation), it will raise the bank rate. This makes borrowing more expensive for commercial banks, who in turn will raise their own lending rates, discouraging borrowing by the public and businesses, thus reducing the money supply.
- **(ii) Open Market Operations (OMO):** This involves the buying and selling of government securities (like bonds) by the Central Bank in the open market. To reduce the money supply, the Central Bank sells securities. Commercial banks and the public buy these securities, and the money they use for the purchase is withdrawn from circulation, reducing the banks' reserves and their ability to lend. To increase the money supply, the Central Bank buys securities, injecting money into the banking system.
- **(iii) Cash Ratio (or Reserve Requirement):** This is the minimum percentage of a commercial bank's total deposits that it is legally required to hold as reserves with the Central Bank. By increasing the cash ratio, the Central Bank forces commercial banks to hold more money in reserve, which reduces the amount of money available for them to lend out, thereby contracting the money supply.

Question 7

(a) Definition of Development Plan:

A development plan is a comprehensive, long-term strategy and a detailed statement of objectives formulated by a government to guide the economic and social development of a country over a specific period. It outlines specific goals (e.g., increasing GDP, reducing poverty, improving infrastructure) and the policies, programs, and resource allocations required to achieve them.

(b) Three problems associated with development planning:

1. **Lack of Reliable Data:** Many developing countries lack the resources to collect accurate, up-to-date statistical data on key economic variables like population, income, and unemployment. Without reliable data, plans are based on guesswork, leading to unrealistic targets and ineffective policies.
2. **Corruption and Mismanagement:** The successful implementation of a plan requires efficient and honest administration. However, corruption can divert public funds meant for development projects into private hands. Mismanagement and bureaucratic inefficiency can also lead to poor execution of projects, causing delays and increasing costs.

3. **Insufficient Funding:** Development plans are often ambitious and require substantial financial resources. Developing countries frequently face challenges in raising sufficient funds, both domestically (due to low savings and a small tax base) and internationally. A lack of capital can cripple the implementation of key projects outlined in the plan.

(c) Three sources of financing a development plan:

1. **Domestic Savings:** This includes savings from individuals, corporations, and the government itself (through budget surpluses). Mobilizing domestic savings through banks and other financial institutions is a crucial and sustainable source of funding for investment in development projects.
2. **Taxation:** Government revenue from direct taxes (like income and corporate tax) and indirect taxes (like VAT and customs duties) is a primary source of finance. An efficient tax system can generate the necessary funds to pay for public infrastructure, education, and healthcare projects outlined in the plan.
3. **Foreign Aid and Loans:** This includes grants (which do not need to be repaid) and loans from foreign governments, international organizations like the World Bank and IMF, and regional development banks. While this can provide significant capital, it can also lead to problems of external debt if not managed properly.

Question 8

(a) Explanation of economic concepts:

These concepts represent different stages of economic integration between countries.

- **(i) Free Trade Area:** This is the first stage of integration where member countries agree to eliminate tariffs, quotas, and other trade barriers among themselves. However, each member country is free to maintain its own independent trade policies (like tariffs) with non-member countries. An example is NAFTA (now USMCA).
- **(ii) Customs Union:** This is a free trade area where member countries not only eliminate trade barriers among themselves but also adopt a common external tariff (CET) and trade policy towards non-member countries. This means they act as a single bloc when trading with the rest of the world.
- **(iii) Common Market:** This stage builds on the customs union by also allowing for the free movement of the factors of production—namely labor and capital—among member countries. This means citizens of one member country can live and work in another, and capital can be invested freely across borders.
- **(iv) Economic Union:** This is a more advanced stage of integration that combines all the features of a common market and also involves the harmonization of national economic policies. This often includes a common currency (monetary union), harmonized tax rates, and common policies on agriculture, industry, and regional development. The European Union (EU) is the most prominent example.

(b) Four problems limiting the effectiveness of ECOWAS:

The Economic Community of West African States (ECOWAS) has faced several challenges that have limited its effectiveness.

1. **Political Instability and Conflicts:** The West African region has historically been prone to political instability, including coups d'état, civil wars, and insurgencies. These conflicts disrupt trade, destroy infrastructure, and force member states to divert resources from economic development to security, undermining regional integration efforts.
2. **Infrastructural Deficiencies:** Poor transportation and communication links between member countries are a major obstacle. Inadequate road networks, railways, and port facilities make the physical movement of goods difficult, costly, and slow, which discourages intra-regional trade.
3. **Overlapping Memberships in other Regional Blocs:** Many ECOWAS member states also belong to other regional economic communities (like the West African Economic and Monetary Union - UEMOA). This creates conflicting obligations, complicates the harmonization of policies (like external tariffs), and divides the loyalty and resources of member states.

4. **Failure to Implement Protocols:** Despite agreeing on numerous protocols (e.g., on free movement of people, goods, and capital), the actual implementation by member states has been slow and inconsistent. National interests often take precedence over regional commitments, and non-tariff barriers (like cumbersome border procedures and corruption) persist, hindering the creation of a truly integrated market.

